Nathan Hunt

nathanhunt94709@gmail.com | linkedin.com/in/nathan-hunt-a6624a30b | nhunt1984.github.io | github.com/nhunt1984

EDUCATION

University of Massachusetts Amherst Anticipated May 2027

Bachelor of Science in Computer Engineering

Relevant coursework: Operating Systems, Computer Architecture, Computer Networks, Network Embedded System Design, Systems Programming, Hardware Organization & Digital Design, Circuits & Electronics I, Junior Design Project

EXPERIENCE

Custom Litt, *Firmware Developer, PixlPro LED panels* May 2026 – Present

- Fixed a memory-overflow crash on legacy hardware by tracing the fault with GDB over JTAG, deferring NVS writes behind a 500 ms cooldown, adding a merge-on-full overflow queue, and closing a double-free (C++, ESP-IDF, FreeRTOS).
- Built the firmware side of HTTPS OTA updates as a host-tested state machine; fixed a bug that wedged it on every app connect.
- Built a pixel-accurate LED simulator by compiling the firmware's own render path for x86 with MinGW, and a Python BLE stress-test harness that flags any write over 50 ms.
- Released firmware 1.0.0 for 8 hardware IDs; adapted a KiCad board for the cable-hider product and generated its JLCPCB files.

UMass CubeSat Team, *Power/Thermals Subteam Member* Sept 2025 – Present

- Sole designer of the satellite's Main Computer, which will command the battery management unit (BMU), control the Power Distribution Unit (PDU) rails, monitor for radiation faults, handle redundancy such as shutting off gates or power cycling, and reflash the flight computer or other processors if they fail; designing it around a radiation-tolerant SAMD21RT with 128 MB of fault-tolerant flash, CAN, USB debug, and six I²C ports.
- Sole designer of a compact 4-layer, 23 × 80 mm KiCad development board around the ATSAMD21J17D-AZ, the COTS drop-in for the SAMD21RT, so the team's programmer can start on firmware before the Main Computer is built; boards are in fabrication.
- Co-designed the PDU's 3.3 V, 5 V, and 12 V rails on TLVM13610 buck modules and a TPS61288 boost, each behind an eFuse for current limiting and software enable; the team's 4-layer, 225-part board was released for fabrication in April 2026.
- Laid out the Main Computer's first test board (ATmega128A) and co-wrote its bare-metal AVR drivers (I²C, telemetry).
- Wrote a 10-page radiation assurance paper for the BMU and PDU, modeling worst-case exposure with AE-8/AP-8, assessing TID, SEE, and DDD, and recommending shielding, rad-tolerant parts, and an analog-preferred circuit approach.

PROJECTS

Plant-Buddy, *BLE soil-sensor probe and Raspberry Pi hub, sole developer* Aug 2026 – Present

- Designed a 4-layer, 30 × 110 mm ESP32-C3 sensor probe in KiCad with 48 SMD parts (FDC1004, SHT40, VEML7700, NTC, EC electrodes, BQ25302 LiPo charger), sourced for JLCPCB assembly; five boards fabricated and brought up.
- Wrote ESP-IDF C firmware (I²C drivers, NimBLE advertising), measuring ~5.4 µA deep sleep against an 11.6 µA budget.
- Caught a soldermask omission over the EC electrodes at bring-up and three broken nets that only a schematic-parity check reveals, then added both checks to the pre-order review.
- Built the hub in Python on a Raspberry Pi CM5: BLE collector into SQLite, calibration pipeline, FastAPI API, and PWA dashboard with Web Push alerts, backed by 982 Python and 262 JavaScript tests.

DIY DJ Deck, *ECE 304 Junior Design Project, two-person team* Feb – May 2026

- Built an audio mixing deck on an STM32F303RE: 44.1 kHz ADC capture, digital mixing with flash-stored samples, and DAC output into a custom KiCad Sallen-Key filter PCB whose cutoff steps from 100 Hz to 12 kHz through PWM-driven optocouplers.

Ai-Tools, *AI agents for KiCad design review, sole developer* Oct 2026 – Present

- Built Claude Code agents that review KiCad schematics, boards, and datasheet constraints on ~15k lines of tested Python tools, validated against KiCad's DRC on 10 boards; they found 12 USB return-path gaps on the Plant-Buddy board.

Operating Systems Projects, *ECE 570/670, C++ on Linux*

- Implemented a user-level thread library from scratch (ucontext context switching, mutexes, condition variables) and a virtual-memory external pager with clock page replacement, verified with stress tests.

BNO055 IMU Driver, *Raspberry Pi Pico, C++17, Pico SDK*

- Wrote a ~940-line I²C driver: full register map, 13 operation modes, axis remap, SI-unit outputs, and calibration restore.

Terminal Text Editor in C

- Wrote a ~630-line editor with termios raw mode and a doubly linked list for O(1) cursor-adjacent edits (~0.05 µs per insert).

SKILLS

Programming: C, C++, Python, TypeScript/JavaScript, Java, ARMv8 Assembly, Verilog, SQL, HTML, CSS
Embedded: ESP-IDF, FreeRTOS, STM32 HAL, RP2040/Pico SDK, SAMD21, AVR, BLE (NimBLE), I²C, SPI, UART, CAN, ADC, DAC, PWM, OTA updates
Hardware: KiCad (4-layer layout, JLCPCB fabrication), LTspice, MATLAB, power electronics (buck/boost regulation, eFuse load switching, battery management), oscilloscope, JTAG/SWD debugging (ST-LINK, GDB), soldering
Software & tools: Git, GitHub Actions, Linux/systemd, FastAPI, SQLite, pytest, PyTorch, Fusion 360, FreeCAD, Claude Code